

CHRISTOPHER VALDIVIA

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PROFESSIONAL SUMMARY

Computer Engineering graduate (Cum Laude, GPA 3.5) with hands-on experience in **software development, embedded systems, networking, cybersecurity, and AI-driven systems**. Proven ability to design, test, and integrate hardware and software solutions using **Python, C/C++, Java, MATLAB, and SQL**. Strong foundation in **system troubleshooting, data analysis, machine learning fundamentals, and secure system design**. Adept at collaborating in cross-functional teams, documenting technical requirements, and delivering reliable, scalable solutions. Actively seeking **Computer Engineering, Software Engineering, Embedded Systems, or IT/Systems Engineering roles**.

CORE TECHNICAL SKILLS

Programming & Software Development

Python, C, C++, Java, MATLAB, SQL, JavaScript, Linux, Bash/Shell scripting, Object-Oriented Programming (OOP), APIs, RESTful services, Git, GitHub, Visual Studio Code

Embedded Systems & Hardware

Microcontrollers (Arduino, Raspberry Pi), Embedded Systems, IoT Systems, Sensor Integration, Circuit Design, PCB Layout, FPGA Fundamentals, RTOS, Edge Computing, Hardware Debugging

Networking & Cybersecurity

Network Configuration, Firewalls, Secure Network Architecture, Ethical Hacking Labs, Incident Response, Vulnerability Assessment, SIEM Tools (Splunk, Microsoft Sentinel), System Hardening

Data, AI & Analytics

Machine Learning Fundamentals, TensorFlow, OpenCV, Data Analytics, Data Visualization (Python, Excel), Power BI, Automation Scripts

Engineering & Development Processes

System Testing, Debugging, Root Cause Analysis, Requirements Analysis, Agile/Scrum, CI/CD Fundamentals, Technical Documentation, Version Control

Tools & Platforms

Microsoft Office (Certified), Adobe Dreamweaver, AutoCAD, Multisim, Proteus, Figma, Jira

EDUCATION

Florida International University (FIU) – Miami, FL

Bachelor of Science in Computer Engineering, *Cum Laude*

GPA: 3.5 | Graduated: August 2025

Barbara Goleman Senior High School – Miami Lakes, FL

High School Diploma, *Cum Laude*

GPA: 3.5 | Graduated: June 2021

CERTIFICATIONS

- Microsoft Security Fundamentals
 - Microsoft Office Specialist (Word, Excel, PowerPoint)
 - Adobe Dreamweaver Certification
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TECHNICAL EXPERIENCE & ACADEMIC PROJECTS

AI-Driven Home Security Enhancement System | Capstone Project – FIU

Jan 2025 – Aug 2025 - Designed and implemented an **AI-powered home security system** leveraging machine learning algorithms, real-time sensor data, and embedded hardware. - Integrated **Python-based automation**, microcontroller programming, and data analytics to improve detection accuracy and system responsiveness. - Collaborated within a **4-member engineering team**, applying Agile principles to debugging, system optimization, and testing, improving overall reliability by **25%**. - Developed detailed **technical documentation**, system diagrams, and formal presentations for engineering faculty review.

Circuit Design & System Integration Projects – FIU

Jan 2024 – Dec 2024 - Designed, simulated, and tested **digital and analog circuits** using Multisim and Proteus. - Performed **hardware-software integration**, validating power efficiency, signal integrity, and system performance. - Conducted **root-cause analysis** and implemented corrective actions during system validation and testing phases.

Cybersecurity Simulation Labs – FIU

Jan 2024 – Dec 2024 - Configured secure network environments for **ethical hacking and defense simulations**. - Implemented **firewalls, authentication mechanisms, and intrusion detection systems**. - Applied cybersecurity best practices for **incident response, vulnerability assessment, and network defense**.

Web Development & Digital Tools – FIU

Jan 2023 – Dec 2023 - Developed responsive web interfaces using **HTML, CSS, JavaScript**, and Adobe Dreamweaver. - Built **data dashboards** using Excel macros and Power BI for reporting and visualization. - Applied **UI/UX principles and accessibility standards** to improve usability and user experience.

ADDITIONAL EXPERIENCE

Volunteer Technical & Educational Support

Barbara Goleman Senior High School / Twin Lakes Elementary School

Jun 2021 – Aug 2025 - Provided **technical support and troubleshooting** for classroom hardware, software, and audiovisual systems. - Assisted faculty with **digital content creation, system setup, and workflow optimization**. - Supported school-wide **technology projects**, improving operational efficiency through software and networking solutions. - Collaborated with staff and administrators, demonstrating strong **communication, adaptability, and reliability**.

LANGUAGES

- English (Fluent)
- Spanish (Fluent)